SAT Math: Hard-Module Challenge 1

Instructions

Complete 12 questions in 30 minutes. A calculator, including a graphing calculator, is permitted. Select exactly one answer (A–D) per question. All choices are exact; do not round intermediate calculations. All variables are real. Use the blank space for work.
Original, unofficial practice targeting harder-module reasoning. This is a challenge set, not a complete adaptive module or a score predictor.

Questions

1. For which value of a does the following system have no solution?

$$(a-2)x+3y=7, \qquad 6x+(a+1)y=14$$

A. -4
B. -2
C. 4
D. 5

2. The quadratic function f is defined by $f(x)=ax^2+bx+c$, where a, b, and c are constants. If $f(2)=24$, $f(4)=-8$, and $f(8)=24$, what is the value of $a+b+c$?

A. -12
B. 4
C. 52
D. 88

3. The function g is defined by $g(x)=ab^x+c$, where $a>0$, $b>1$, and c is a constant. The table gives three values of g.

x	0	2	4
$g(x)$	11	35	131

What is $g(1)$?

A. 17
B. 19
C. 23
D. 47

4. A factory uses only production lines A and B. In one day's production, 4% of the components from line A and 12% of the components from line B were defective. Overall, 7% of that day's components were defective. If a component is selected at random from that day's nondefective components, what is the probability that it came from line B?

   A. $\frac{7}{100}$
   B. $\frac{3}{25}$
   C. $\frac{3}{8}$
   D. $\frac{11}{31}$

5. For each constant k, consider the equation

$$\frac{x^2 - 9}{x - 3} = kx.$$

   What is the sum of all values of k for which the equation has no solution?

   A. 1
   B. 2
   C. 3
   D. 4

6. For each constant k, consider the equation

$$(k - 3)x^2 + 12x + 3 = 0.$$

   What is the sum of all values of k for which the equation has exactly one real solution?

   A. 3
   B. 12
   C. 15
   D. 18

7. A theater predicts that it will sell 240 tickets to a performance if each ticket costs \$30. For every \$3 increase in the ticket price, the theater predicts that it will sell 12 fewer tickets. The price may be increased only in \$3 increments. The cost of presenting the performance is \$1,800 plus \$6 for each ticket sold. Under this model, what is the greatest predicted profit, in dollars, from the performance?

   A. 3,960
   B. 5,220
   C. 5,256
   D. 8,064

8. Two right circular cylinders are similar. The ratio of the total surface area of the smaller cylinder to that of the larger cylinder is 9 : 25. The larger cylinder's volume is 784 cubic centimeters greater than the smaller cylinder's volume. What is the sum of their volumes, in cubic centimeters?

   A. 980
   B. 1,216
   C. 1,568
   D. 3,136

9. In the xy-plane, the line $y = 2x + 1$ is tangent to the circle

$$x^2 + y^2 - 10x + 6y + c = 0,$$

   where c is a constant. What is the value of c?

   A. $-\frac{196}{5}$
   B. $-\frac{26}{5}$
   C. $\frac{26}{5}$
   D. $\frac{196}{5}$

10. In right triangle ABC, angle C is a right angle. Point D lies on hypotenuse AB, and CD is perpendicular to AB. If $AD : DB = 4 : 9$ and the area of triangle ABC is 156 square units, what is the perimeter of triangle ABC, in units?

A. $26 + 10\sqrt{13}$
B. $26 + 20\sqrt{13}$
C. $52 + 10\sqrt{13}$
D. $52 + 20\sqrt{13}$

11. For which positive value of k does the equation

$$\left|x^2 - 8x + 7\right| = k$$

have exactly three distinct real solutions?

A. 7
B. 8
C. 9
D. 16

12. If x satisfies

$$2^x + 2^{-x} = 5,$$

what is the value of $2^{2x} + 2^{-2x}$?

A. 23
B. 25
C. 27
D. 625